



PSoC® Creator™

Project Datasheet for

AcondicionamientoAnalogico

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1 Overview

The Cypress PSoC 5 is a family of 32-bit devices with the following characteristics:

- High-performance 32-bit ARM Cortex-M3 core with a nested vectored interrupt controller (NVIC) and a high-performance DMA controller
- Digital system that includes configurable Universal Digital Blocks (UDBs) and specific function peripherals, such as USB, I2C and SPI
- Analog subsystem that includes 20-bit Delta Sigma converters (ADC), SAR ADCs, 8-bit DACs that can be configured for 12-bit operation, comparators, op amps and configurable switched capacitor (SC) and continuous time (CT) blocks to create PGAs, TIAs, mixers, and more
- Several types of memory elements, including SRAM, flash, and EEPROM
- Programming and debug system through JTAG, serial wire debug (SWD), and single wire viewer (SWV)
- Flexible routing to all pins

Figure 1 shows the major components of a typical [CY8C58LP](#) series member PSoC 5LP device. For details on all the systems listed above, please refer to the [PSoC 5LP Technical Reference Manual](#).

Figure 1. CY8C58LP Device Series Block Diagram

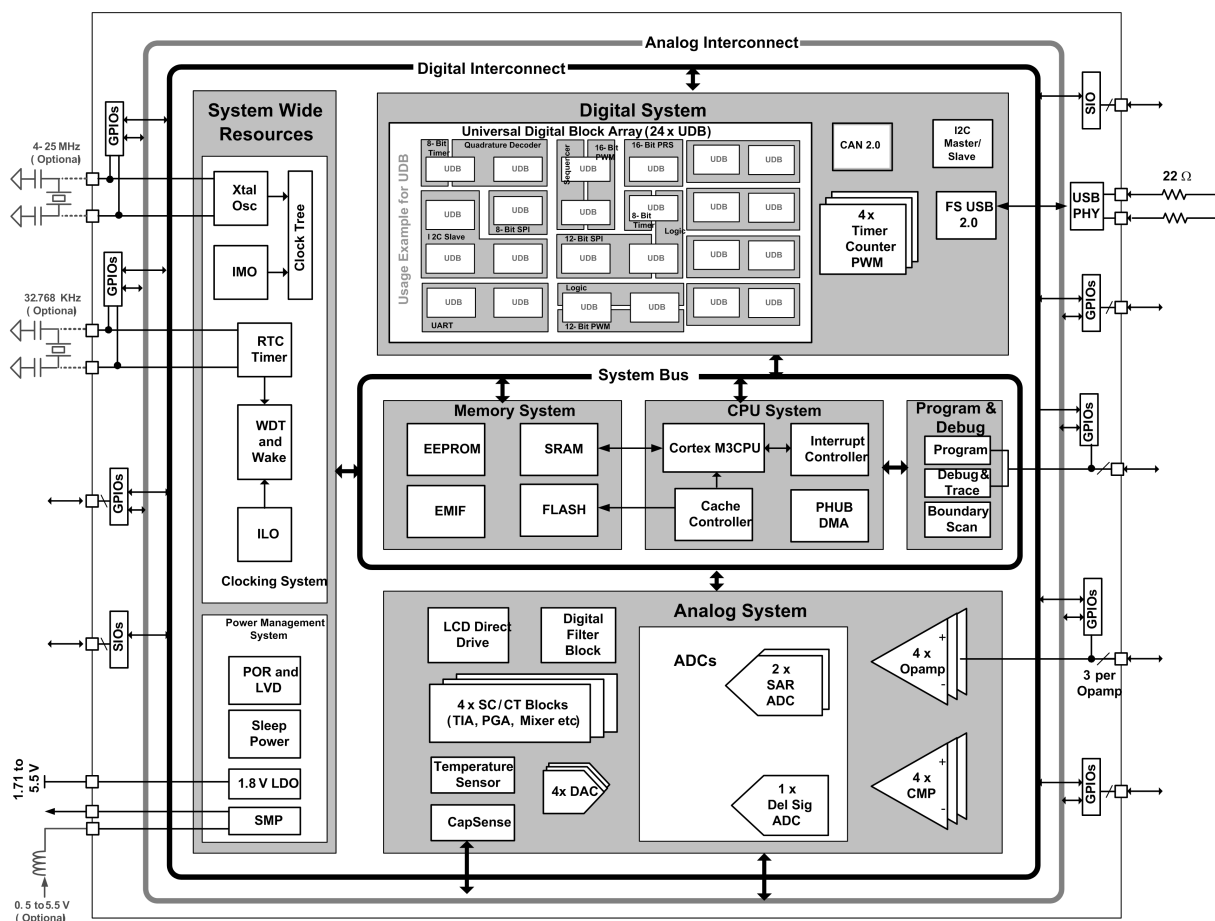


Table 1 lists the key characteristics of this device.

Table 1. Device Characteristics

Name	Value
Part Number	CY8C5888LTI-LP097
Package Name	68-QFN
Family	PSoC 5LP
Series	CY8C58LP
Max CPU speed (MHz)	0
Flash size (kB)	256
SRAM size (kB)	64
EEPROM size (bytes)	2048
Vdd range (V)	1.71 to 5.5
Automotive qualified	No (Industrial Grade Only)
Temp range (Celsius)	-40 to 85
JTAG ID	0x2E161069

NOTE: The CPU speed noted above is the maximum available speed. The CPU is clocked by Bus Clock, listed in the [System Clocks](#) section below.

Table 2 lists the device resources that this design uses:

Table 2. Device Resources

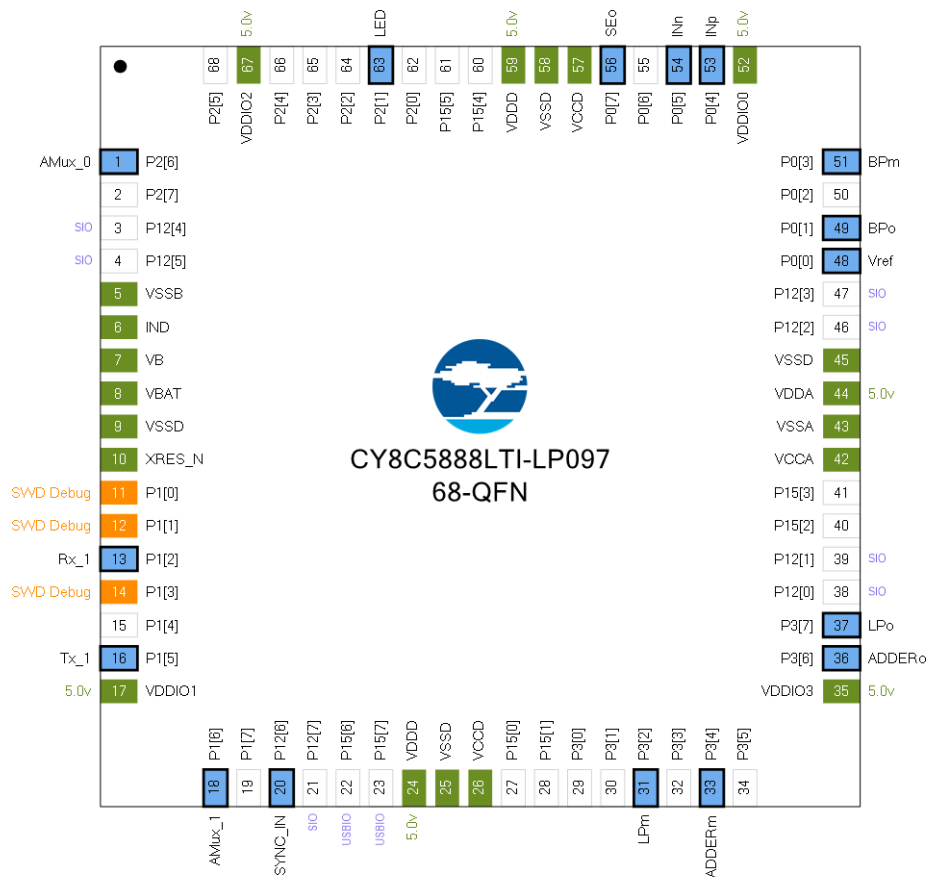
Resource Type	Used	Free	Max	% Used
Digital Clocks	2	6	8	25.00 %
Analog Clocks	1	3	4	25.00 %
CapSense Buffers	0	2	2	0.00 %
Digital Filter Block	0	1	1	0.00 %
Interrupts	6	26	32	18.75 %
IO	19	29	48	39.58 %
Segment LCD	0	1	1	0.00 %
CAN 2.0b	0	1	1	0.00 %
I2C	0	1	1	0.00 %
USB	0	1	1	0.00 %
DMA Channels	0	24	24	0.00 %
Timer	0	4	4	0.00 %
UDB				
Macrocells	32	160	192	16.67 %
Unique P-terms	53	331	384	13.80 %
Total P-terms	63			
Datapath Cells	7	17	24	29.17 %
Status Cells	5	19	24	20.83 %
StatusI Registers	3			
Sync Cells (x3)	1			
Routed Count7 Load/Enable	1			
Control Cells	2	22	24	8.33 %
Control Registers	1			
Count7 Cells	1			
Opamp	4	0	4	100.00 %
Comparator	0	4	4	0.00 %
Delta-Sigma ADC	1	0	1	100.00 %
LPF	2	0	2	100.00 %
SAR ADC	0	2	2	0.00 %
Analog (SC/CT) Blocks	3	1	4	75.00 %

Resource Type	Used	Free	Max	% Used
DAC				
VIDAC	4	0	4	100.00 %

2 Pins

Figure 2 shows the pin layout of this device.

Figure 2. Device Pin Layout



2.1 Hardware Pins

Table 3 contains information about the pins on this device in device pin order. (No connection ["n/c"] pins have been omitted.)

Table 3. Device Pins

Pin	Port	Name	Type	Drive Mode	Reset State
1	P2[6]	AMux_0	Analog	HiZ analog	HiZ Analog Unb
2	P2[7]	GPIO [unused]			HiZ Analog Unb
3	P12[4]	SIO [unused]			HiZ Analog Unb
4	P12[5]	SIO [unused]			HiZ Analog Unb
5	VSSB	VSSB	Dedicated		
6	IND	IND	Dedicated		
7	VB	VB	Dedicated		
8	VBAT	VBAT	Dedicated		
9	VSSD	VSSD	Power		
10	XRES_N	XRES_N	Dedicated		
11	P1[0]	Debug:SWD_IO	Reserved		
12	P1[1]	Debug:SWD_CK	Reserved		
13	P1[2]	Rx_1	Dgtl In	HiZ digital	HiZ Analog Unb
14	P1[3]	Debug:SWV	Reserved		
15	P1[4]	GPIO [unused]			HiZ Analog Unb
16	P1[5]	Tx_1	Dgtl Out	OD, DL	HiZ Analog Unb
17	VDDIO1	VDDIO1	Power		
18	P1[6]	AMux_1	Analog	HiZ analog	HiZ Analog Unb
19	P1[7]	GPIO [unused]			HiZ Analog Unb
20	P12[6]	SYNC_IN	Dgtl In	HiZ digital	HiZ Analog Unb
21	P12[7]	SIO [unused]			HiZ Analog Unb
22	P15[6]	USB IO [unused]			HiZ Analog Unb
23	P15[7]	USB IO [unused]			HiZ Analog Unb
24	VDDD	VDDD	Power		
25	VSSD	VSSD	Power		
26	VCCD	VCCD	Power		
27	P15[0]	GPIO [unused]			HiZ Analog Unb
28	P15[1]	GPIO [unused]			HiZ Analog Unb
29	P3[0]	GPIO [unused]			HiZ Analog Unb
30	P3[1]	GPIO [unused]			HiZ Analog Unb
31	P3[2]	LPm	Analog	HiZ analog	HiZ Analog Unb
32	P3[3]	GPIO [unused]			HiZ Analog Unb
33	P3[4]	ADDERm	Analog	HiZ analog	HiZ Analog Unb
34	P3[5]	GPIO [unused]			HiZ Analog Unb
35	VDDIO3	VDDIO3	Power		
36	P3[6]	ADDERo	Analog	HiZ analog	HiZ Analog Unb
37	P3[7]	LPo	Analog	HiZ analog	HiZ Analog Unb
38	P12[0]	SIO [unused]			HiZ Analog Unb
39	P12[1]	SIO [unused]			HiZ Analog Unb
40	P15[2]	GPIO [unused]			HiZ Analog Unb
41	P15[3]	GPIO [unused]			HiZ Analog Unb
42	VCCA	VCCA	Power		
43	VSSA	VSSA	Power		
44	VDDA	VDDA	Power		
45	VSSD	VSSD	Power		

Pin	Port	Name	Type	Drive Mode	Reset State
46	P12[2]	SIO [unused]			HiZ Analog Unb
47	P12[3]	SIO [unused]			HiZ Analog Unb
48	P0[0]	Vref	Analog	HiZ analog	HiZ Analog Unb
49	P0[1]	BPo	Analog	HiZ analog	HiZ Analog Unb
50	P0[2]	GPIO [unused]			HiZ Analog Unb
51	P0[3]	BPm	Analog	HiZ analog	HiZ Analog Unb
52	VDDIO0	VDDIO0	Power		
53	P0[4]	INp	Analog	HiZ analog	HiZ Analog Unb
54	P0[5]	INn	Analog	HiZ analog	HiZ Analog Unb
55	P0[6]	GPIO [unused]			HiZ Analog Unb
56	P0[7]	SEo	Analog	HiZ analog	HiZ Analog Unb
57	VCCD	VCCD	Power		
58	VSSD	VSSD	Power		
59	VDDD	VDDD	Power		
60	P15[4]	GPIO [unused]			HiZ Analog Unb
61	P15[5]	GPIO [unused]			HiZ Analog Unb
62	P2[0]	GPIO [unused]			HiZ Analog Unb
63	P2[1]	LED	Software In/Out	Strong drive	HiZ Analog Unb
64	P2[2]	GPIO [unused]			HiZ Analog Unb
65	P2[3]	GPIO [unused]			HiZ Analog Unb
66	P2[4]	GPIO [unused]			HiZ Analog Unb
67	VDDIO2	VDDIO2	Power		
68	P2[5]	GPIO [unused]			HiZ Analog Unb

Abbreviations used in Table 3 have the following meanings:

- HiZ analog = High impedance analog
- HiZ Analog Unb = Hi-Z Analog Unbuffered
- Dgtl In = Digital Input
- HiZ digital = High impedance digital
- Dgtl Out = Digital Output
- OD, DL = Open drain, drives low

2.2 Hardware Ports

Table 4 contains information about the pins on this device in device port order. (No connection ["n/c"], power and dedicated pins have been omitted.)

Table 4. Device Ports

Port	Pin	Name	Type	Drive Mode	Reset State
P0[0]	48	Vref	Analog	HiZ analog	HiZ Analog Unb
P0[1]	49	BPo	Analog	HiZ analog	HiZ Analog Unb
P0[2]	50	GPIO [unused]			HiZ Analog Unb
P0[3]	51	BPm	Analog	HiZ analog	HiZ Analog Unb
P0[4]	53	INp	Analog	HiZ analog	HiZ Analog Unb
P0[5]	54	INn	Analog	HiZ analog	HiZ Analog Unb
P0[6]	55	GPIO [unused]			HiZ Analog Unb
P0[7]	56	SEo	Analog	HiZ analog	HiZ Analog Unb
P1[0]	11	Debug:SWD_IO	Reserved		
P1[1]	12	Debug:SWD_CK	Reserved		
P1[2]	13	Rx_1	Dgtl In	HiZ digital	HiZ Analog Unb
P1[3]	14	Debug:SWV	Reserved		
P1[4]	15	GPIO [unused]			HiZ Analog Unb
P1[5]	16	Tx_1	Dgtl Out	OD, DL	HiZ Analog Unb
P1[6]	18	AMux_1	Analog	HiZ analog	HiZ Analog Unb
P1[7]	19	GPIO [unused]			HiZ Analog Unb
P12[0]	38	SIO [unused]			HiZ Analog Unb
P12[1]	39	SIO [unused]			HiZ Analog Unb
P12[2]	46	SIO [unused]			HiZ Analog Unb
P12[3]	47	SIO [unused]			HiZ Analog Unb
P12[4]	3	SIO [unused]			HiZ Analog Unb
P12[5]	4	SIO [unused]			HiZ Analog Unb
P12[6]	20	SYNC_IN	Dgtl In	HiZ digital	HiZ Analog Unb
P12[7]	21	SIO [unused]			HiZ Analog Unb
P15[0]	27	GPIO [unused]			HiZ Analog Unb
P15[1]	28	GPIO [unused]			HiZ Analog Unb
P15[2]	40	GPIO [unused]			HiZ Analog Unb
P15[3]	41	GPIO [unused]			HiZ Analog Unb
P15[4]	60	GPIO [unused]			HiZ Analog Unb
P15[5]	61	GPIO [unused]			HiZ Analog Unb
P15[6]	22	USB IO [unused]			HiZ Analog Unb
P15[7]	23	USB IO [unused]			HiZ Analog Unb
P2[0]	62	GPIO [unused]			HiZ Analog Unb
P2[1]	63	LED	Software In/Out	Strong drive	HiZ Analog Unb
P2[2]	64	GPIO [unused]			HiZ Analog Unb
P2[3]	65	GPIO [unused]			HiZ Analog Unb
P2[4]	66	GPIO [unused]			HiZ Analog Unb
P2[5]	68	GPIO [unused]			HiZ Analog Unb
P2[6]	1	AMux_0	Analog	HiZ analog	HiZ Analog Unb
P2[7]	2	GPIO [unused]			HiZ Analog Unb
P3[0]	29	GPIO [unused]			HiZ Analog Unb
P3[1]	30	GPIO [unused]			HiZ Analog Unb
P3[2]	31	LPm	Analog	HiZ analog	HiZ Analog Unb
P3[3]	32	GPIO [unused]			HiZ Analog Unb
P3[4]	33	ADDERm	Analog	HiZ analog	HiZ Analog Unb

Port	Pin	Name	Type	Drive Mode	Reset State
P3[5]	34	GPIO [unused]			HiZ Analog Unb
P3[6]	36	ADDERo	Analog	HiZ analog	HiZ Analog Unb
P3[7]	37	LPo	Analog	HiZ analog	HiZ Analog Unb

Abbreviations used in Table 4 have the following meanings:

- HiZ analog = High impedance analog
- HiZ Analog Unb = Hi-Z Analog Unbuffered
- Dgtl In = Digital Input
- HiZ digital = High impedance digital
- Dgtl Out = Digital Output
- OD, DL = Open drain, drives low

2.3 Software Pins

Table 5 contains information about the software pins on this device in alphabetical order. (Only software-accessible pins are shown.)

Table 5. Software Pins

Name	Port	Type	Reset State
ADDERm	P3[4]	Analog	HiZ Analog Unb
ADDERo	P3[6]	Analog	HiZ Analog Unb
AMux_0	P2[6]	Analog	HiZ Analog Unb
AMux_1	P1[6]	Analog	HiZ Analog Unb
BPm	P0[3]	Analog	HiZ Analog Unb
BPo	P0[1]	Analog	HiZ Analog Unb
Debug:SWD_CK	P1[1]	Reserved	
Debug:SWD_IO	P1[0]	Reserved	
Debug:SWV	P1[3]	Reserved	
GPIO [unused]	P2[4]		HiZ Analog Unb
GPIO [unused]	P15[3]		HiZ Analog Unb
GPIO [unused]	P15[2]		HiZ Analog Unb
GPIO [unused]	P3[3]		HiZ Analog Unb
GPIO [unused]	P3[1]		HiZ Analog Unb
GPIO [unused]	P3[5]		HiZ Analog Unb
GPIO [unused]	P2[5]		HiZ Analog Unb
GPIO [unused]	P15[4]		HiZ Analog Unb
GPIO [unused]	P15[5]		HiZ Analog Unb
GPIO [unused]	P2[0]		HiZ Analog Unb
GPIO [unused]	P0[6]		HiZ Analog Unb
GPIO [unused]	P2[3]		HiZ Analog Unb
GPIO [unused]	P0[2]		HiZ Analog Unb
GPIO [unused]	P2[2]		HiZ Analog Unb
GPIO [unused]	P3[0]		HiZ Analog Unb
GPIO [unused]	P2[7]		HiZ Analog Unb
GPIO [unused]	P1[7]		HiZ Analog Unb
GPIO [unused]	P1[4]		HiZ Analog Unb
GPIO [unused]	P15[1]		HiZ Analog Unb
GPIO [unused]	P15[0]		HiZ Analog Unb
INn	P0[5]	Analog	HiZ Analog Unb
INp	P0[4]	Analog	HiZ Analog Unb
LED	P2[1]	Software In/Out	HiZ Analog Unb
LPm	P3[2]	Analog	HiZ Analog Unb
LPo	P3[7]	Analog	HiZ Analog Unb
Rx_1	P1[2]	Dgtl In	HiZ Analog Unb
SEo	P0[7]	Analog	HiZ Analog Unb
SIO [unused]	P12[4]		HiZ Analog Unb
SIO [unused]	P12[5]		HiZ Analog Unb
SIO [unused]	P12[2]		HiZ Analog Unb
SIO [unused]	P12[1]		HiZ Analog Unb
SIO [unused]	P12[0]		HiZ Analog Unb
SIO [unused]	P12[3]		HiZ Analog Unb
SIO [unused]	P12[7]		HiZ Analog Unb
SYNC_IN	P12[6]	Dgtl In	HiZ Analog Unb
Tx_1	P1[5]	Dgtl Out	HiZ Analog Unb

Name	Port	Type	Reset State
USB IO [unused]	P15[6]		HiZ Analog Unb
USB IO [unused]	P15[7]		HiZ Analog Unb
Vref	P0[0]	Analog	HiZ Analog Unb

Abbreviations used in Table 5 have the following meanings:

- HiZ Analog Unb = Hi-Z Analog Unbuffered
- Dgtl In = Digital Input
- Dgtl Out = Digital Output

For more information on reading, writing and configuring pins, please refer to:

- Pins chapter in the [System Reference Guide](#)
 - o CyPins API routines
- Programming Application Interface section in the [cy_pins component datasheet](#)

3 System Settings

3.1 System Configuration

Table 6. System Configuration Settings

Name	Value
Device Configuration Mode	Compressed
Enable Error Correcting Code (ECC)	False
Store Configuration Data in ECC Memory	True
Instruction Cache Enabled	True
Enable Fast IMO During Startup	True
Unused Bonded IO	Allow but warn
Heap Size (bytes)	0x80
Stack Size (bytes)	0x0800
Include CMSIS Core Peripheral Library Files	True

3.2 System Debug Settings

Table 7. System Debug Settings

Name	Value
Debug Select	SWD+SWV (serial wire debug and viewer)
Enable Device Protection	False
Embedded Trace (ETM)	False
Use Optional XRES	False

3.3 System Operating Conditions

Table 8. System Operating Conditions

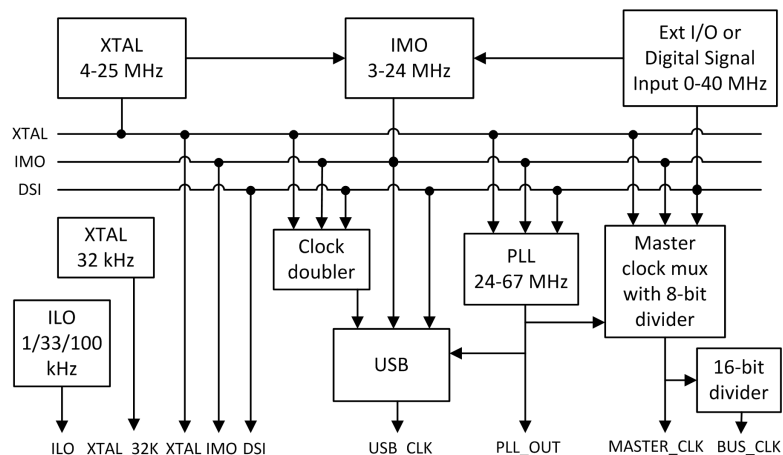
Name	Value
VDDA (V)	5.0
VDDD (V)	5.0
VDDIO0 (V)	5.0
VDDIO1 (V)	5.0
VDDIO2 (V)	5.0
VDDIO3 (V)	5.0
Variable VDDA	False
Temperature Range	-40C - 85/125C

4 Clocks

The clock system includes these clock resources:

- Four internal clock sources increase system integration:
 - o 3 to 74.7 MHz Internal Main Oscillator (IMO) $\pm 1\%$ at 3 MHz
 - o 1 kHz, 33 kHz, and 100 kHz Internal Low Speed Oscillator (ILO) outputs
 - o 12 to 80 MHz clock doubler output, sourced from IMO, MHz External Crystal Oscillator (MHzECO), and Digital System Interconnect (DSI)
 - o 24 to 80 MHz fractional Phase-Locked Loop (PLL) sourced from IMO, MHzECO, and DSI
- Clock generated using a DSI signal from an external I/O pin or other logic
- Two external clock sources provide high precision clocks:
 - o 4 to 25 MHz External Crystal Oscillator (MHzECO)
 - o 32.768 kHz External Crystal Oscillator (kHzECO) for Real Time Clock (RTC)
- Dedicated 16-bit divider for bus clock
- Eight individually sourced 16-bit clock dividers for the digital system peripherals
- Four individually sourced 16-bit clock dividers with skew for the analog system peripherals
- IMO has a USB mode that synchronizes to USB host traffic, requiring no external crystal for USB. (USB equipped parts only)

Figure 3. System Clock Configuration



4.1 System Clocks

Table 9 lists the system clocks used in this design.

Table 9. System Clocks

Name	Domain	Source	Desired Freq	Nominal Freq	Accuracy (%)	Start at Reset	Enabled
BUS_CLK	DIGITAL	MASTER_CLK	? MHz	24 MHz	±1	True	True
PLL_OUT	DIGITAL	IMO	24 MHz	24 MHz	±1	True	True
MASTER_CLK	DIGITAL	PLL_OUT	? MHz	24 MHz	±1	True	True
IMO	DIGITAL		3 MHz	3 MHz	±1	True	True
ILO	DIGITAL		? MHz	100 kHz	-55,+100	True	True
USB_CLK	DIGITAL	IMO	48 MHz	? MHz	±0	False	False
XTAL	DIGITAL		24 MHz	? MHz	±0	False	False
XTAL 32kHz	DIGITAL		32.768 kHz	? MHz	±0	False	False
Digital Signal	DIGITAL		? MHz	? MHz	±0	False	False

4.2 Local and Design Wide Clocks

Local clocks drive individual analog and digital blocks. Design wide clocks are a user-defined optimization, where two or more analog or digital blocks that share a common clock profile (frequency, etc) can be driven from the same clock divider output source.

Figure 4. Local and Design Wide Clock Configuration

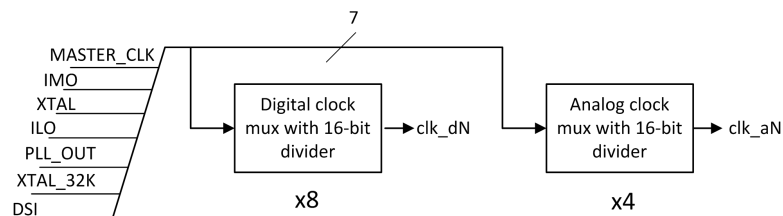


Table 10 lists the local clocks used in this design.

Table 10. Local Clocks

Name	Domain	Source	Desired Freq	Nominal Freq	Accuracy (%)	Start at Reset	Enabled
ADC_Ext_CP_-Clk	DIGITAL	MASTER_CLK	? MHz	24 MHz	±1	True	True
timer_clock_1	DIGITAL	BUS_CLK	? MHz	24 MHz	±1	True	True
ADC_theACLK	ANALOG	MASTER_CLK	3.072 MHz	3 MHz	±1	True	True
UART_IntClock	DIGITAL	MASTER_CLK	921.6 kHz	923.077 kHz	±1	True	True
timer_clock	DIGITAL	ILO	? MHz	100 kHz	-55,+100	True	True

For more information on clocking resources, please refer to:

- Clocking System chapter in the [PSoC 5LP Technical Reference Manual](#)
- Clocking chapter in the [System Reference Guide](#)
 - o CyPLL API routines
 - o CyIMO API routines

4 Clocks



- o CyILO API routines
- o CyMaster API routines
- o CyXTAL API routines

5 Interrupts and DMAs

5.1 Interrupts

This design contains the following interrupt components: (0 is the highest priority)

Table 11. Interrupts

Name	Intr Num	Vector	Priority
UART_RXInternalInterrupt	0	0	7
UART_TXInternalInterrupt	1	1	7
isr_DelSig	2	2	7
isr_SyncIn	3	3	7
isr_Timer	4	4	7
ADC_IRQ	29	29	7

For more information on interrupts, please refer to:

- Interrupt Controller chapter in the [PSoC 5LP Technical Reference Manual](#)
- Interrupts chapter in the [System Reference Guide](#)
 - o CyInt API routines and related registers
- Datasheet for [cy_isr component](#)

5.2 DMAs

This design contains no DMA components.

6 Flash Memory

PSoC 5LP devices offer a host of Flash protection options and device security features that you can leverage to meet the security and protection requirements of an application. These requirements range from protecting configuration settings or Flash data to locking the entire device from external access.

Table 12 lists the Flash protection settings for your design.

Table 12. Flash Protection Settings

Start Address	End Address	Protection Level
0x0	0x3FFFF	U - Unprotected

Flash memory is organized as rows with each row of flash having 256 bytes. Each flash row can be assigned one of four protection levels:

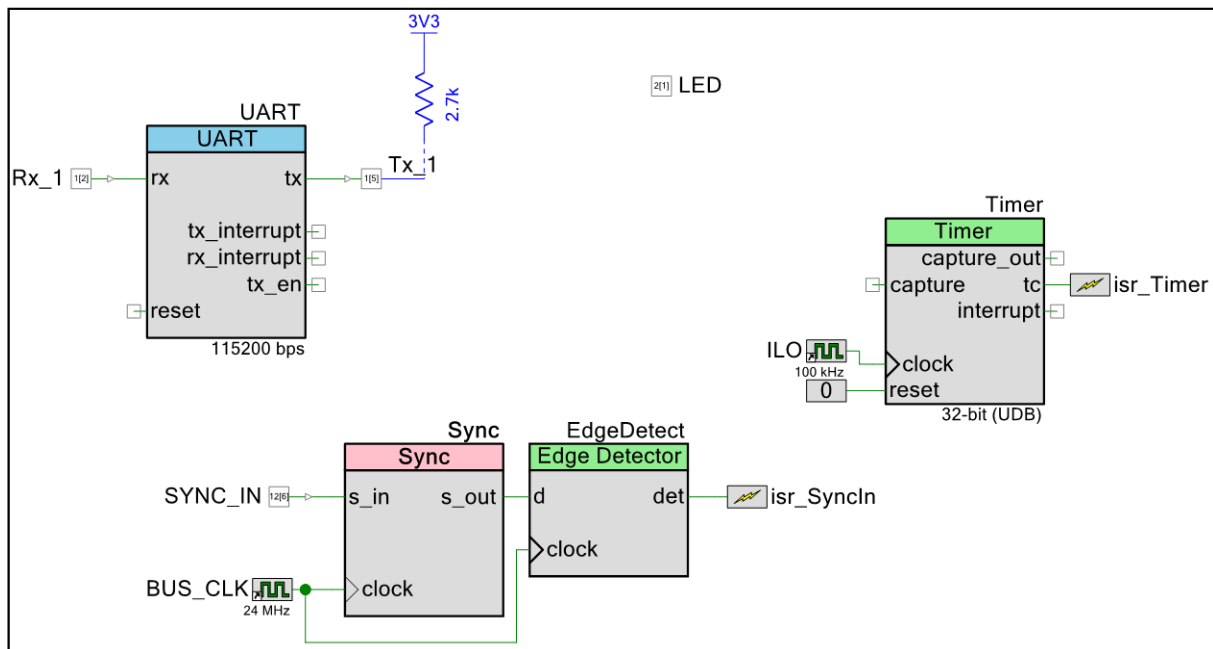
- U - Unprotected
- F - Factory Upgrade
- R - Field Upgrade
- W - Full Protection

For more information on Flash memory and protection, please refer to:

- Flash Protection chapter in the [PSoC 5LP Technical Reference Manual](#)
- Flash and EEPROM chapter in the [System Reference Guide](#)
 - o CyWrite API routines
 - o CyFlash API routines

7.2 Schematic Sheet: Digital

Figure 6. Schematic Sheet: Digital



This schematic sheet contains the following component instances:

- Instance [EdgeDetect](#) (type: EdgeDetect_v1_0)
- Instance [Sync](#) (type: cy_sync_v1_0)
- Instance [Timer](#) (type: Timer_v2_80)
- Instance [UART](#) (type: UART_v2_50)

8 Components

8.1 Component type: ADC_DelSig [v3.30]

8.1.1 Instance ADC

Description: Delta-Sigma ADC

Instance type: ADC_DelSig [v3.30]

Datasheet: [online component datasheet for ADC_DelSig](#)

Table 13. Component Parameters for ADC

Parameter Name	Value	Description
ADC_Alignment	Left OVF Protected	This parameter determines how the result is aligned in the 24 bit result word.
ADC_Alignment_Config2	Right	This parameter determines how the result is aligned in the 24 bit result word.
ADC_Alignment_Config3	Right	This parameter determines how the result is aligned in the 24 bit result word.
ADC_Alignment_Config4	Right	This parameter determines how the result is aligned in the 24 bit result word.
ADC_Charge_Pump_Clock	true	Low power charge pump clock selection
ADC_Clock	Internal	Parameter for selecting the ADC clock type.
ADC_Input_Mode	Differential	Differential or Single ended input mode
ADC_Input_Range	-Input +/- 2*Vref	Choose input operating mode that best supports the range of the signals being measured.
ADC_Input_Range_Config2	-Input +/- 2*Vref	Choose input operating mode that best supports the range of the signals being measured.
ADC_Input_Range_Config3	-Input +/- Vref	Choose input operating mode that best supports the range of the signals being measured.
ADC_Input_Range_Config4	-Input +/- Vref	Choose input operating mode that best supports the range of the signals being measured.
ADC_Power	Medium Power	Sets power level of ADC.
ADC_Reference	Internal Vdda/4	Selects voltage reference source and configuration.
ADC_Reference_Config2	Internal Vdda/4	Selects voltage reference source and configuration.
ADC_Reference_Config3	Internal 1.024 Volts	Selects voltage reference source and configuration.
ADC_Reference_Config4	Internal 1.024 Volts	Selects voltage reference source and configuration.
ADC_Resolution	18	ADC Resolution in bits
ADC_Resolution_Config2	18	ADC Resolution in bits
ADC_Resolution_Config3	16	ADC Resolution in bits
ADC_Resolution_Config4	16	ADC Resolution in bits

Parameter Name	Value	Description
Clock_Frequency	64000	Determines the ADC clock frequency.
Comment_Config1	Default Config	Parameter which holds the user comment for the config1.
Comment_Config2	Second Config	Parameter which holds the user comment for the config2.
Comment_Config3	Third Config	Parameter which holds the user comment for the config3.
Comment_Config4	Fourth Config	Parameter which holds the user comment for the config4.
Config1_Name	CFG1	This parameter is used to create constants in the header file for config 1.
Config2_Name	CFG2	This parameter is used to create constants in the header file for config 2.
Config3_Name	CFG3	This parameter is used to create constants in the header file for config 3.
Config4_Name	CFG4	This parameter is used to create constants in the header file for config 4.
Configs	2	Number of active configurations
Conversion_Mode	2 - Continuous	ADC conversion mode
Conversion_Mode_Config2	0 - Single Sample	ADC conversion mode
Conversion_Mode_Config3	2 - Continuous	ADC conversion mode
Conversion_Mode_Config4	2 - Continuous	ADC conversion mode
Enable_Vref_Vss	false	Determines whether or not to connect ADC's reference Vssa to AGL[6].
EnableModulatorInput	false	When this parameter is enabled, the modulator input terminal will be enabled on the symbol.
Input_Buffer_Gain	1	Gain of input amplifier
Input_Buffer_Gain_Config2	1	Gain of input amplifier
Input_Buffer_Gain_Config3	1	Gain of input amplifier
Input_Buffer_Gain_Config4	1	Gain of input amplifier
Input_Buffer_Mode	Bypass Buffer	Buffer Mode type selection
Input_Buffer_Mode_Config2	Bypass Buffer	Buffer Mode type selection
Input_Buffer_Mode_Config3	Rail to Rail	Buffer Mode type selection
Input_Buffer_Mode_Config4	Rail to Rail	Buffer Mode type selection
Ref_Voltage	1.25	Set reference voltage
Ref_Voltage_Config2	1.25	Set reference voltage
Ref_Voltage_Config3	1.024	Set reference voltage
Ref_Voltage_Config4	1.024	Set reference voltage
rm_int	false	Removes internal interrupt (IRQ)
Sample_Rate	3000	Sample Rate in Hz
Sample_Rate_Config2	2475	Sample Rate in Hz
Sample_Rate_Config3	10000	Sample Rate in Hz
Sample_Rate_Config4	10000	Sample Rate in Hz
Start_of_Conversion	Software	Continuous conversions or hardware controlled
User Comments		Instance-specific comments.

8.2 Component type: AMux [v1.80]

8.2.1 Instance AMux_ADC

Description: Multiplexer used to route analog signals.

Instance type: AMux [v1.80]

Datasheet: [online component datasheet for AMux](#)

Table 14. Component Parameters for AMux_ADC

Parameter Name	Value	Description
AtMostOneActive	true	Limit to at most one active channel.
Channels	4	Channel count.
Isolation	Maximum	Specify minimum, medium, or maximum switch control; affects channel isolation and switching time.
MuxType	Single	Select between single or differential inputs.
User Comments		Instance-specific comments.

8.2.2 Instance AMux_IN

Description: Multiplexer used to route analog signals.

Instance type: AMux [v1.80]

Datasheet: [online component datasheet for AMux](#)

Table 15. Component Parameters for AMux_IN

Parameter Name	Value	Description
AtMostOneActive	true	Limit to at most one active channel.
Channels	2	Channel count.
Isolation	Maximum	Specify minimum, medium, or maximum switch control; affects channel isolation and switching time.
MuxType	Differential	Select between single or differential inputs.
User Comments		Instance-specific comments.

8.3 Component type: Analog_LPF [v1.0]

8.3.1 Instance LPF_1

Description: Analog Low Pass Filter

Instance type: Analog_LPF [v1.0]

Datasheet: (not available)

Table 16. Component Parameters for LPF_1

Parameter Name	Value	Description
Cutoff_Frequency	15 kHz	
User Comments		Instance-specific comments.

8.3.2 Instance LPF_2

Description: Analog Low Pass Filter

Instance type: Analog_LPF [v1.0]

Datasheet: (not available)

Table 17. Component Parameters for LPF_2

Parameter Name	Value	Description
Cutoff_Frequency	15 kHz	
User Comments		Instance-specific comments.

8.4 Component type: cy_sync [v1.0]

8.4.1 Instance Sync

Description: Synchronizes the input signal(s) to its input clock

Instance type: cy_sync [v1.0]

Datasheet: [online component datasheet for cy_sync](#)

Table 18. Component Parameters for Sync

Parameter Name	Value	Description
SignalWidth	1	This parameter configures the number of signals that will be synchronized to the associated clock.
User Comments		Instance-specific comments.

8.5 Component type: EdgeDetect [v1.0]

8.5.1 Instance EdgeDetect

Description: Edge Detector

Instance type: EdgeDetect [v1.0]

Datasheet: [online component datasheet for EdgeDetect](#)

Table 19. Component Parameters for EdgeDetect

Parameter Name	Value	Description
EdgeType	Either Edge	The edge type that this component will detect.
User Comments		Instance-specific comments.

8.6 Component type: OpAmp [v1.90]

8.6.1 Instance OPAadder

Description: Opamp

Instance type: OpAmp [v1.90]

Datasheet: [online component datasheet for OpAmp](#)

Table 20. Component Parameters for OPAadder

Parameter Name	Value	Description
Mode	OpAmp	Selects between uncommitted op-amp or follower mode.
Power	High Power	Selects the device power level.

Parameter Name	Value	Description
User Comments		Instance-specific comments.

8.6.2 Instance OPAbp

Description: Opamp

Instance type: OpAmp [v1.90]

Datasheet: [online component datasheet for OpAmp](#)

Table 21. Component Parameters for OPAbp

Parameter Name	Value	Description
Mode	OpAmp	Selects between uncommitted op-amp or follower mode.
Power	High Power	Selects the device power level.
User Comments		Instance-specific comments.

8.6.3 Instance OPAlp

Description: Opamp

Instance type: OpAmp [v1.90]

Datasheet: [online component datasheet for OpAmp](#)

Table 22. Component Parameters for OPAlp

Parameter Name	Value	Description
Mode	OpAmp	Selects between uncommitted op-amp or follower mode.
Power	High Power	Selects the device power level.
User Comments		Instance-specific comments.

8.6.4 Instance OPAref

Description: Opamp

Instance type: OpAmp [v1.90]

Datasheet: [online component datasheet for OpAmp](#)

Table 23. Component Parameters for OPAref

Parameter Name	Value	Description
Mode	Follower	Selects between uncommitted op-amp or follower mode.
Power	High Power	Selects the device power level.
User Comments		Instance-specific comments.

8.7 Component type: PGA [v2.0]

8.7.1 Instance PGAGain

Description: Programmable Gain Amplifier

Instance type: PGA [v2.0]

Datasheet: [online component datasheet for PGA](#)

Table 24. Component Parameters for PGAGain

Parameter Name	Value	Description
Gain	50	Selects supported gain value.

Parameter Name	Value	Description
Power	High Power	Selects the device power.
User Comments		Instance-specific comments.
Vref_Input	External	Enables direct connection from the Analog ground (Agnd) to the inverting input.

8.7.2 Instance PGAn

Description: Programmable Gain Amplifier

Instance type: PGA [v2.0]

Datasheet: [online component datasheet for PGA](#)

Table 25. Component Parameters for PGAn

Parameter Name	Value	Description
Gain	2	Selects supported gain value.
Power	High Power	Selects the device power.
User Comments		Instance-specific comments.
Vref_Input	External	Enables direct connection from the Analog ground (Agnd) to the inverting input.

8.7.3 Instance PGAp

Description: Programmable Gain Amplifier

Instance type: PGA [v2.0]

Datasheet: [online component datasheet for PGA](#)

Table 26. Component Parameters for PGAp

Parameter Name	Value	Description
Gain	2	Selects supported gain value.
Power	High Power	Selects the device power.
User Comments		Instance-specific comments.
Vref_Input	External	Enables direct connection from the Analog ground (Agnd) to the inverting input.

8.8 Component type: Timer [v2.80]

8.8.1 Instance Timer

Description: 8, 16, 24 or 32-bit Timer

Instance type: Timer [v2.80]

Datasheet: [online component datasheet for Timer](#)

Table 27. Component Parameters for Timer

Parameter Name	Value	Description
CaptureAlternatingFall	false	Enables data capture on either edge but not until a valid falling edge is detected first.
CaptureAlternatingRise	false	Enables data capture on either edge but not until a valid rising edge is detected first.

Parameter Name	Value	Description
CaptureCount	2	The CaptureCount parameter works as a divider on the hardware input "capture". A CaptureCount value of 2 would result in an actual capture taking place every other time the input "capture" is changed.
CaptureCounterEnabled	false	Enables the capture counter to count capture events (up to 127) before a capture is triggered.
CaptureMode	Rising Edge	This parameter defines the capture input signal requirements to trigger a valid capture event
EnableMode	Software Only	This parameter specifies the methods in enabling the component. Hardware mode makes the enable input pin visible. Software mode may reduce the resource usage if not enabled.
FixedFunction	false	Configures the component to use fixed function HW block instead of the UDB implementation.
InterruptOnCapture	false	Parameter to check whether interrupt on a capture event is enabled or disabled.
InterruptOnFIFOFull	false	Parameter to check whether interrupt on a FIFO Full event is enabled disabled.
InterruptOnTC	false	Parameter to check whether interrupt on a TC is enabled or disabled.
NumberOfCaptures	1	Number of captures allowed until the counter is cleared or disabled.
Period	255	Defines the timer period (This is also the reload value when terminal count is reached)
Resolution	32	Defines the resolution of the hardware. This parameter affects how many bits are used in the Period counter and defines the maximum resolution of the internal component signals.
RunMode	One Shot	Defines the hardware to run continuously, run until a terminal count is reached or run until an interrupt event is triggered.
TriggerMode	None	Defines the required trigger input signal to cause a valid trigger enable of the timer
User Comments		Instance-specific comments.

8.9 Component type: UART [v2.50]

8.9.1 Instance UART

Description: Universal Asynchronous Receiver Transmitter

Instance type: UART [v2.50]

Datasheet: [online component datasheet for UART](#)

Table 28. Component Parameters for UART

Parameter Name	Value	Description
Address1	0	This parameter specifies the RX Hardware Address #1.
Address2	0	This parameter specifies the RX Hardware Address #2.
BaudRate	115200	Sets the target baud rate.
BreakBitsRX	13	Specifies the break signal length for the RX (detection) channel.
BreakBitsTX	13	Specifies the break signal length for the TX channel.
BreakDetect	false	Enables the break detect hardware.
CRCOutputsEn	false	Enables the CRC outputs.
EnIntRXInterrupt	true	Enables the internal RX interrupt configuration and the ISR.
EnIntTXInterrupt	true	Enables the internal TX interrupt configuration and the ISR.
FlowControl	None	Enable the flow control signals.
HalfDuplexEn	false	Enables half duplex mode on the RX Half of the UART module.
HwTXEnSignal	true	Enables the external TX enable signal output.
InternalClock	true	Enables the internal clock. This parameter removes the clock input pin.
InterruptOnTXComplete	false	This is an Interrupt mask used to enable/disable the interrupt on 'TX complete' event.
InterruptOnTXFifoEmpty	true	This is an Interrupt mask used to enable/disable the interrupt on 'TX FIFO empty' event.
InterruptOnTXFifoFull	false	This is an Interrupt mask used to enable/disable the interrupt on 'TX FIFO full' event.
InterruptOnTXFifoNotFull	false	This is an Interrupt mask used to enable/disable the interrupt on 'TX FIFO not full' event.
IntOnAddressDetect	false	Enables the interrupt on hardware address detected event by default
IntOnAddressMatch	false	Enables the interrupt on hardware address match detected event by default
IntOnBreak	false	Enables the interrupt on break signal detected event by default
IntOnByteRcvd	true	Enables the interrupt on RX byte received event by default

Parameter Name	Value	Description
IntOnOverrunError	false	Enables the interrupt on overrun error event by default
IntOnParityError	false	Enables the interrupt on parity error event by default
IntOnStopError	false	Enables the interrupt on stop error event by default
NumDataBits	8	Defines the number of data bits. Values can be 5, 6, 7 or 8 bits.
NumStopBits	1	Defines the number of stop bits. Values can be 1 or 2 bits.
OverSamplingRate	8	This parameter defines the over sampling rate.
ParityType	None	Sets the parity type as Odd, Even or Mark/Space
ParityTypeSw	false	This parameter allows the parity type to be changed through software by using the WriteControlRegister API
RXAddressMode	None	Configures the RX hardware address detection mode
RXBufferSize	32	The size of the RAM space allocated for the RX input buffer.
RXEnable	true	Enables the RX in the UART
TXBitClkGenDP	true	When enabled, this parameter enables the TX clock generation on DataPath resource. When disabled, TX clock is generated from Clock7.
TXBufferSize	128	The size of the RAM space allocated for the TX output buffer.
TXEnable	true	Enables the TX in the UART
Use23Polling	true	Allows the use of 2 out of 3 polling resources on the RX UART sampler.
User Comments		Instance-specific comments.

8.10 Component type: VDAC8 [v1.90]

8.10.1 Instance VDAC_Ref_Adder

Description: 8-Bit Voltage DAC

Instance type: VDAC8 [v1.90]

Datasheet: [online component datasheet for VDAC8](#)

Table 29. Component Parameters for VDAC_Ref_Adder

Parameter Name	Value	Description
Data_Source	CPU or DMA (Data Bus)	Selects the method in which the data is written to the vDAC.
Initial_Value	255	Configures the initial vDAC output voltage. The output uses the following relation: Initial output voltage = value*(FullRange/255). This calculated output voltage value is invalid if DAC Bus is used.

Parameter Name	Value	Description
Strobe_Mode	Register Write	Selects how the data is strobed into the DAC. For a register write, the data is strobed into the DAC on each CPU or DMA write. If operating in External mode, an external data strobe signal is required.
User Comments		Instance-specific comments.
VDAC_Range	0 - 1.020V (4mV/bit)	Specifies the full voltage scale range of the vDAC
VDAC_Speed	High Speed	Specifies the vDAC settling speed. Note that the 'Slow Speed' selection consumes less power.
Voltage	1020	This parameter sets the voltage value.

8.10.2 Instance VDAC_ref_BP

Description: 8-Bit Voltage DAC

Instance type: VDAC8 [v1.90]

Datasheet: [online component datasheet for VDAC8](#)

Table 30. Component Parameters for VDAC_ref_BP

Parameter Name	Value	Description
Data_Source	CPU or DMA (Data Bus)	Selects the method in which the data is written to the vDAC.
Initial_Value	255	Configures the initial vDAC output voltage. The output uses the following relation: Initial output voltage = value*(FullRange/255). This calculated output voltage value is invalid if DAC Bus is used.
Strobe_Mode	Register Write	Selects how the data is strobed into the DAC. For a register write, the data is strobed into the DAC on each CPU or DMA write. If operating in External mode, an external data strobe signal is required.
User Comments		Instance-specific comments.
VDAC_Range	0 - 1.020V (4mV/bit)	Specifies the full voltage scale range of the vDAC
VDAC_Speed	High Speed	Specifies the vDAC settling speed. Note that the 'Slow Speed' selection consumes less power.
Voltage	1020	This parameter sets the voltage value.

8.10.3 Instance VDAC_ref_LP

Description: 8-Bit Voltage DAC

Instance type: VDAC8 [v1.90]

Datasheet: [online component datasheet for VDAC8](#)

Table 31. Component Parameters for VDAC_ref_LP

Parameter Name	Value	Description
Data_Source	CPU or DMA (Data Bus)	Selects the method in which the data is written to the vDAC.
Initial_Value	255	Configures the initial vDAC output voltage. The output uses the following relation: Initial output voltage = $\text{value} * (\text{FullRange} / 255)$. This calculated output voltage value is invalid if DAC Bus is used.
Strobe_Mode	Register Write	Selects how the data is strobed into the DAC. For a register write, the data is strobed into the DAC on each CPU or DMA write. If operating in External mode, an external data strobe signal is required.
User Comments		Instance-specific comments.
VDAC_Range	0 - 1.020V (4mV/bit)	Specifies the full voltage scale range of the vDAC
VDAC_Speed	High Speed	Specifies the vDAC settling speed. Note that the 'Slow Speed' selection consumes less power.
Voltage	1020	This parameter sets the voltage value.

8.10.4 Instance VDAC_ref_PGA

Description: 8-Bit Voltage DAC

Instance type: VDAC8 [v1.90]

Datasheet: [online component datasheet for VDAC8](#)

Table 32. Component Parameters for VDAC_ref_PGA

Parameter Name	Value	Description
Data_Source	CPU or DMA (Data Bus)	Selects the method in which the data is written to the vDAC.
Initial_Value	255	Configures the initial vDAC output voltage. The output uses the following relation: Initial output voltage = $\text{value} * (\text{FullRange} / 255)$. This calculated output voltage value is invalid if DAC Bus is used.
Strobe_Mode	Register Write	Selects how the data is strobed into the DAC. For a register write, the data is strobed into the DAC on each CPU or DMA write. If operating in External mode, an external data strobe signal is required.
User Comments		Instance-specific comments.
VDAC_Range	0 - 1.020V (4mV/bit)	Specifies the full voltage scale range of the vDAC

Parameter Name	Value	Description
VDAC_Speed	High Speed	Specifies the vDAC settling speed. Note that the 'Slow Speed' selection consumes less power.
Voltage	1020	This parameter sets the voltage value.

9 Other Resources

The following documents contain important information on Cypress software APIs that might be relevant to this design:

- Standard Types and Defines chapter in the [System Reference Guide](#)
 - o Software base types
 - o Hardware register types
 - o Compiler defines
 - o Cypress API return codes
 - o Interrupt types and macros
- Registers
 - o The full PSoC 5LP register map is covered in the [PSoC 5LP Registers Technical Reference Manual](#)
 - o Register Access chapter in the [System Reference Guide](#)
 - § CY_GET API routines
 - § CY_SET API routines
- System Functions chapter in the [System Reference Guide](#)
 - o General API routines
 - o CyDelay API routines
 - o CyVd Voltage Detect API routines
- Power Management
 - o Power Supply and Monitoring chapter in the [PSoC 5LP Technical Reference Manual](#)
 - o Low Power Modes chapter in the [PSoC 5LP Technical Reference Manual](#)
 - o Power Management chapter in the [System Reference Guide](#)
 - § CyPm API routines
- Watchdog Timer chapter in the [System Reference Guide](#)
 - o CyWdt API routines
- Cache Management
 - o Cache Controller chapter in the [PSoC 5LP Technical Reference Manual](#)
 - o Cache chapter in the [System Reference Guide](#)
 - § CyFlushCache() API routine